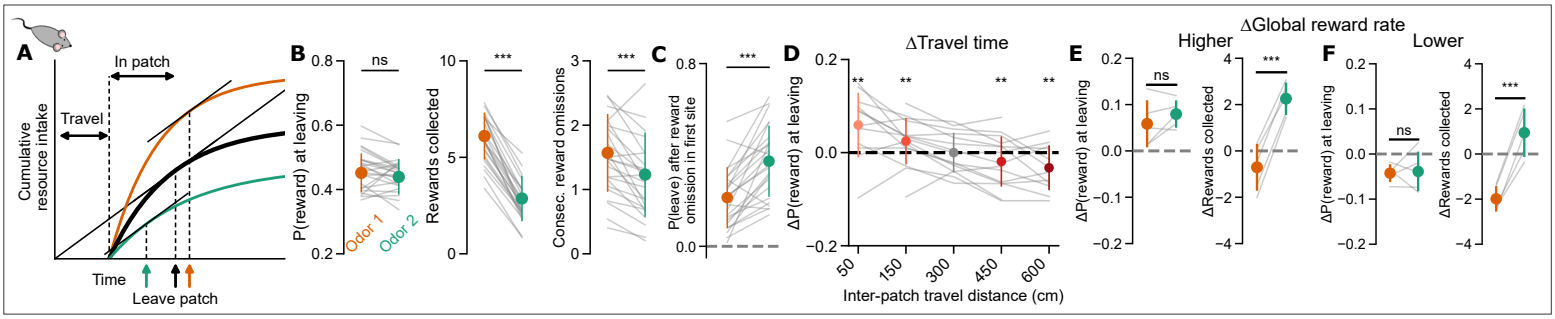
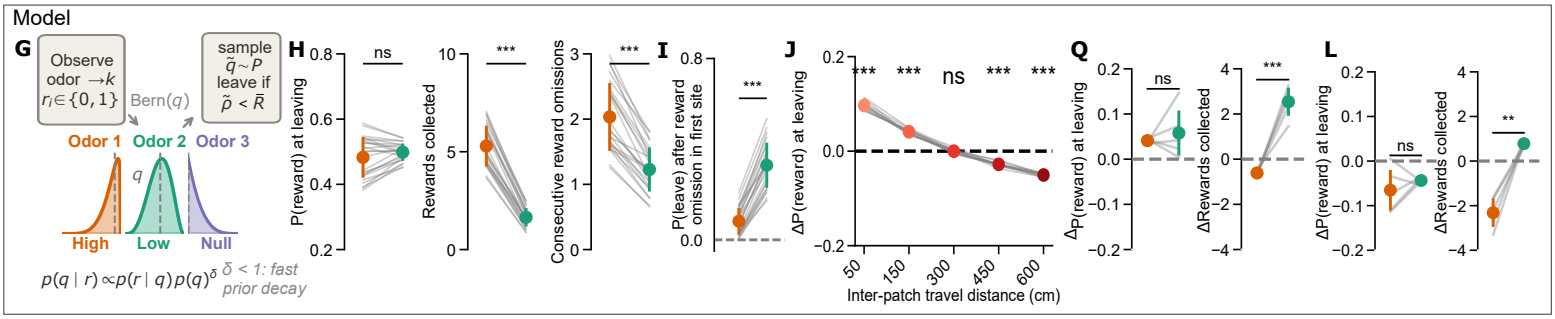

A
In patch
Travel
Cumulative resource intake
Time
Leave patch
B
P(reward) at leaving
Odor 1
Odor 2
ns
Rewards collected
***
Consec. reward omissions
C
P(leave) after reward omission in first site
D
ΔTravel time
ΔP(reward) at leaving
Inter-patch travel distance (cm)
E
ΔGlobal reward rate
Higher
ΔP(reward) at leaving
ΔRewards collected
F
Lower


Model
G
Observe odor →k
r_i ∈ {0, 1}
Bern(q)
sample q̃ ~ P
leave if p̃ < R̄
Odor 1
Odor 2
Odor 3
q
High
Low
Null
p(q | r) ∝ p(r | q) p(q)^δ
δ < 1: fast prior decay
H
P(reward) at leaving
ns
Rewards collected
***
Consecutive reward omissions
I
P(leave) after reward omission in first site
J
ΔP(reward) at leaving
Inter-patch travel distance (cm)
Q
ΔP(reward) at leaving
ΔRewards collected
L